3. Match list I with list II

List I	List II
A. A pair of chromosomes extra with diploid	i) monosomy
B. One chromosome extra to the diploid	ii) tetrasomy
C. One chromosome loses from diploid	iii) trisomy
D. Two individual chromosomes lose from diploid	iv) double monosomy

a) A-i, B-iii, C-ii, D-iv b) A-ii, B-iii, C-iv, D-i
c) A-ii, B-iii, C-i, D-iv d) A-iii, B-ii, C-i, D-iv

4. Which of the following sentences are correct?
   1. The offspring exhibit only parental combinations due to incomplete linkage
   2. The linked genes exhibit some crossing over in complete linkage
   3. The separation of two linked genes are possible in incomplete linkage
   4. Crossing over is absent in complete linkage

   a) 1 and 2 b) 2 and 3
   c) 3 and 4 d) 1 and 4

5. Accurate mapping of genes can be done by three point test cross because increases
   a) Possibility of single cross over
   b) Possibility of double cross over
   c) Possibility of multiple cross over
   d) Possibility of recombination frequency

6. Due to incomplete linkage in maize, the ratio of parental and recombinants are
   a) 50:50 b) 7:1:1:7 c) 96.4: 3.6 d) 1:7:7:1

7. Genes **G S L H** are located on same chromosome. The recombination percentage is between L and G is 15%, S and L is 50%, H and S are 20%. The correct order of genes is
   a) GHSL b) SHGL c) SGHL d) HSLG

8. The point mutation sequence for transition, transition, transversion and transversion in DNA are
   a) A to T, T to A, C to G and G to C
   b) A to G, C to T, C to G and T to A
   c) C to G, A to G, T to A and G to A
   d) G to C, A to T, T to A and C to G

9. If haploid number in a cell is 18. The double monosomic and trisomic number will be
   a) 35 and 37 b) 34 and 35
   c) 37 and 35 d) 17 and 19

10. Changing the codon AGC to AGA represents
    a) missense mutation b) nonsense mutation
    c) frameshift mutation d) deletion mutation

11. **Assertion (A):** Gamma rays are generally use to induce mutation in wheat varieties.
    Reason (R): Because they carry lower energy to non-ionize electrons from atom
    a) A is correct. R is correct explanation of A
    b) A is correct. R is not correct explanation of A
    c) A is correct. R is wrong explanation of A
    d) A and R is wrong

12. How many map units separate two alleles A and B if the recombination frequency is 0.09?
    a) 900 cM b) 90 cM c) 9 cM d) 0.9 cM

13. When two different genes came from same parent they tend to remain together.
    i) What is the name of this phenomenon?
    ii) Draw the cross with suitable example.
    iii) Write the observed phenotypic ratio.

14. If you cross dominant genotype PV/PV male *Drosophila* with double recessive female and obtain F_1 hybrid. Now you cross

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